Module 2
Practical No 6-A
Menus, Dialogs and UI Interaction

Aim: Develop an Android application implementing AppBar (Toolbar) and Options Menu.

Menu

- In Android, the menu is a vital UI element that is used to offer certain common functionality throughout the application.
- With the help of the menu, users can experience a smooth and consistent experience throughout the application.
- Menu needs to be defined in a separate XML file, which we should then use in our programme in accordance with our needs.

AppBar (Toolbar)

The **Toolbar** is a modern and flexible version of the ActionBar. It appears at the top of the screen and displays the application title, navigation buttons, and menu items.

Options Menu

The **Options Menu** provides a collection of actions or commands for the user. It appears when the user clicks the menu icon on the Toolbar.

Note: 1) To create menu.xml file Right click on res–Click on new resource file==set file name=mymenu and resource type =menu

1) activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

<androidx.appcompat.widget.Toolbar
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:id="@+id/toolbar"
  android:background="#3F51B5"
  app:layout_constraintEnd_toEndOf="parent"
```

```
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu Practical"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

2) mymenu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item android:id="@+id/search"
        android:title="Search"/>
    <item android:id="@+id/share"
        android:title="Share"/>
    <item android:id="@+id/app_bar_delete"
        android:title="Delete"/>
    <item android:id="@+id/app_bar_save"
        android:title="Save"/>
    <item android:id="@+id/app_bar_exit"
        android:title="Exit"
        />

</menu>
```

3) MainActivity.java

```
package com.example.menu_pract;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.widget.Toolbar;
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

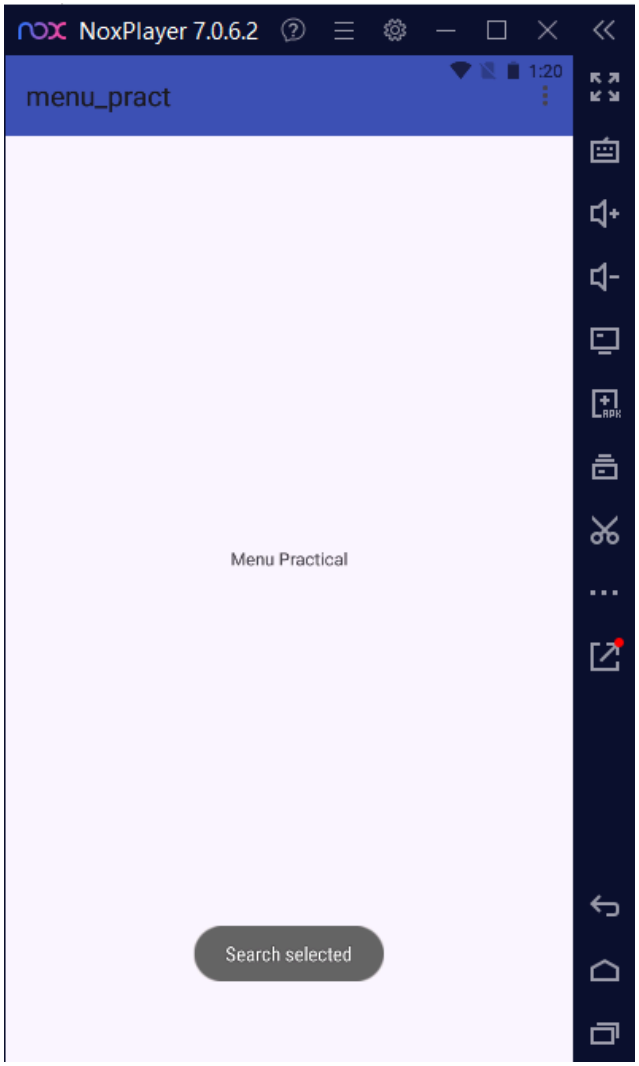
public class MainActivity extends AppCompatActivity {
  Toolbar t1;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);
     t1 = findViewById(R.id.toolbar);
     setSupportActionBar(t1);
  }
  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
     MenuInflater menuInflater = new MenuInflater(this);
     menuInflater.inflate(R.menu.mymenu,menu);
     return true;
  }

  @Override
  public boolean onOptionsItemSelected(@NonNull MenuItem item) {
     int id = item.getItemId();
     if (id == R.id.search) {
        Toast.makeText(this, "Search selected", Toast.LENGTH_SHORT).show();
        return true;
     }
     else if (id == R.id.app_bar_save) {
        Toast.makeText(this, "Save selected", Toast.LENGTH_SHORT).show();
        return true;
     }
     else if (id == R.id.share) {
        Toast.makeText(this, "Share selected", Toast.LENGTH_SHORT).show();
        return true;
     }
     else if (id == R.id.app_bar_exit) {
        Toast.makeText(this, "Exit selected", Toast.LENGTH_SHORT).show();
        return true;
     }
     else if (id == R.id.app_bar_delete) {
        Toast.makeText(this, "Delete selected", Toast.LENGTH_SHORT).show();
        return true;
```

```
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output:



Note: import this class if getting error in **setSupportActionBar(t1);**

```
import androidx.appcompat.widget.Toolbar;
```
And remove existing toolbar class

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